Statistical learning and deep learning

How machines learn from examples

POLI 158: AI & Politics · Session 3

Yiqing Xu

Picking up from Session 1

The big picture

AI	algorithms and systems that mimic human reasoning and decisions
Machine learning (ML)	AI methods that learn rules or patterns directly from data
Deep learning (DL)	machine learning with neural networks of many layers
Generative AI (GenAI)	models that learn what the data look like, to create new content
Large language models (LLMs)	generative AI built on transformers and trained on huge amounts of text

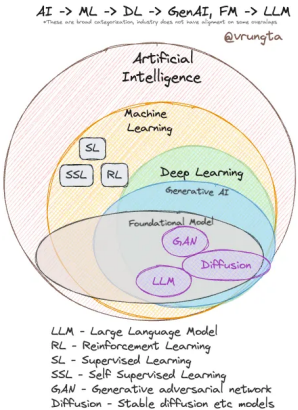


Figure: @vrungta (vrungta.substack.com).

AI and deep learning by task (examples)

Natural language processing
understand, interpret, and generate human language

Computer vision
understand images and videos

Robotics
design, build, and run robots that do tasks

Speech recognition
turn speech into text, or text into speech

Science
analyze data and make discoveries across fields

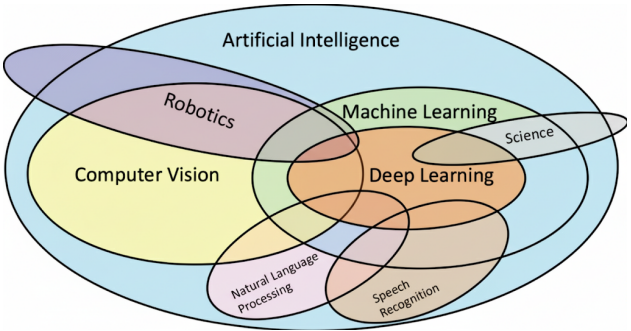
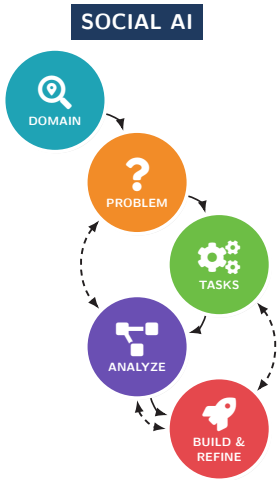


Figure: Yuan Yuan (Boston College, CSCI 3370, 2024), after Fei-Fei Li and Justin Johnson.

Social and political applications of AI

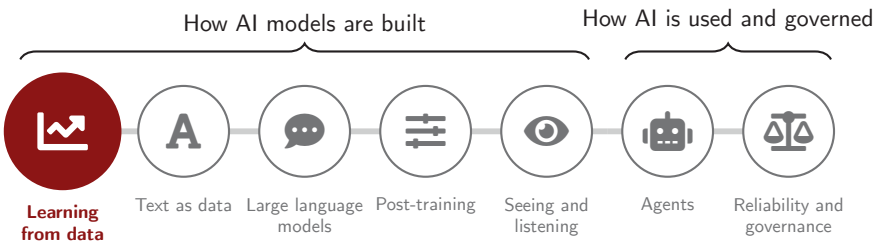
1. Find your **domain**: what area interests you most?
2. Locate the **problem**: what bothers people in this domain? Is it scalable? How large is the "market"?
3. Define the **tasks**: what should AI solve?
4. Analyze the technical challenges: which **AI architecture** should you use? Ask Claude or Codex.
5. **Build**, and set up a fast feedback loop.



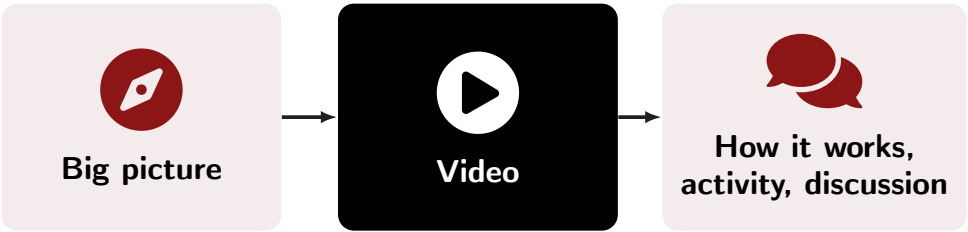
TODAY'S QUESTION

Is a model that predicts well good enough to use in policymaking?

The seven Wednesday sessions



Today in three parts



Where machine learning is used

Everyday life

- spam filters
- face unlock
- recommendations
- translation

Research

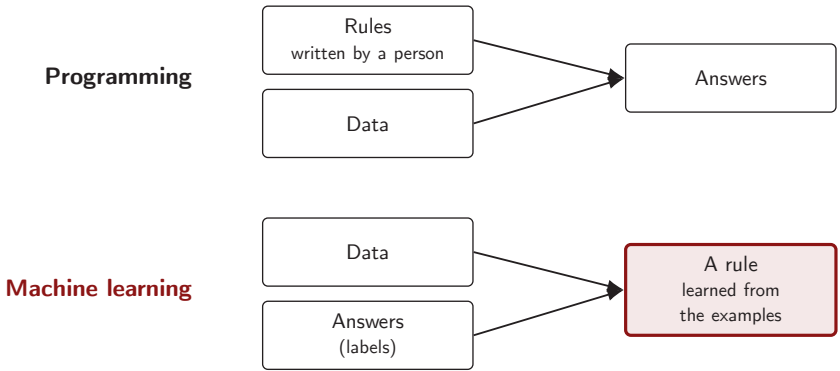
- poverty maps
- protest events

Jean et al. (2016); Zhang and Pan (2019)

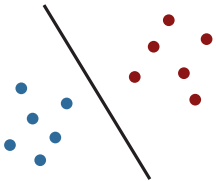
Policymaking

- risk scores in court
- benefit fraud flags
- police flags

Programming and machine learning



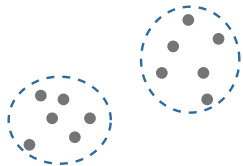
Three kinds of learning



Supervised

examples with answers

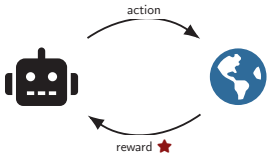
Will this bill pass?



Unsupervised

examples without answers

Which legislators vote alike?

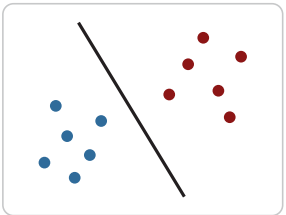


Reinforcement

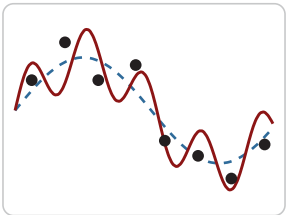
trial, error, and rewards

Chatbots trained on feedback

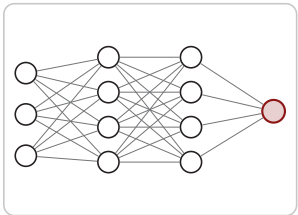
Three concepts in today's video



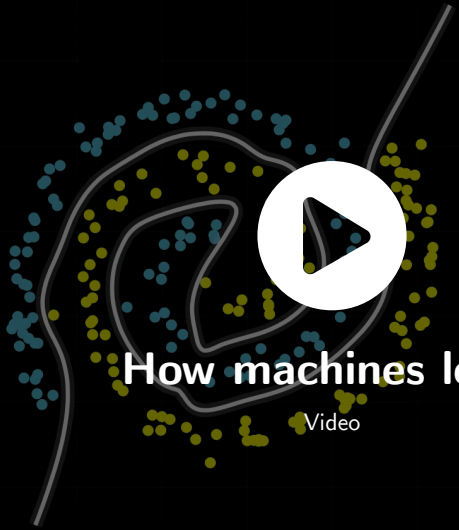
Learning from examples



Overfitting



Neural networks



How machines learn

Video

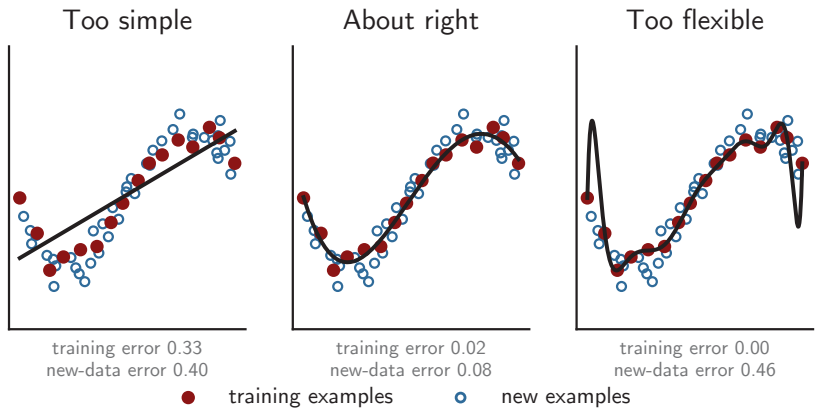
passes through the data

100

dots on the right side

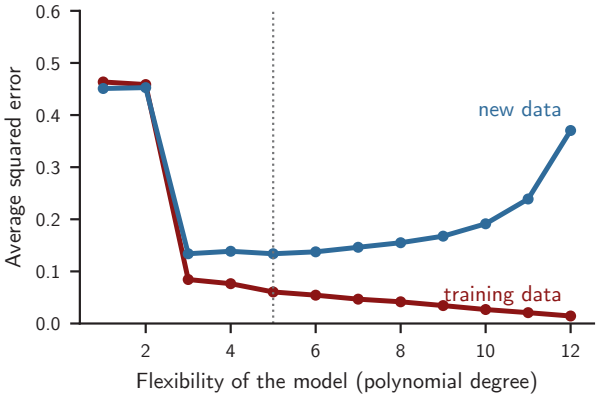
100%

Fits with three levels of flexibility



Simulated data: a smooth curve plus noise. Error = average squared distance to the points.

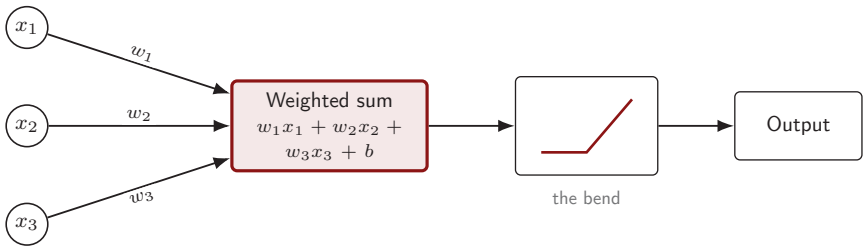
Training error and new-data error



- Dotted line: the flexibility with the lowest error on new data.

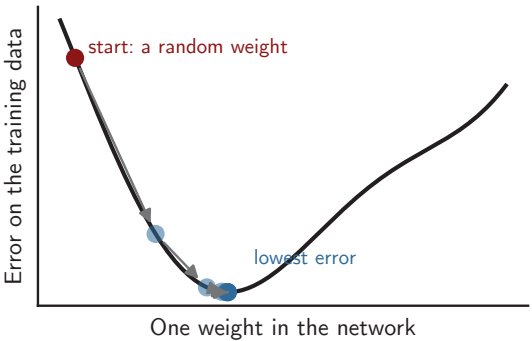
Same simulation as the last slide, repeated 300 times. The curves show the median error.

A neuron



- The weights w are the numbers the network learns.

Gradient descent



1. Start with random weights.
2. Measure the error on a batch of examples.
3. Change every weight a little to lower the error.
4. Repeat, millions of times.

Deep learning since 2012



Data

ImageNet: about 14 million labeled images



Computing

Graphics chips (GPUs)



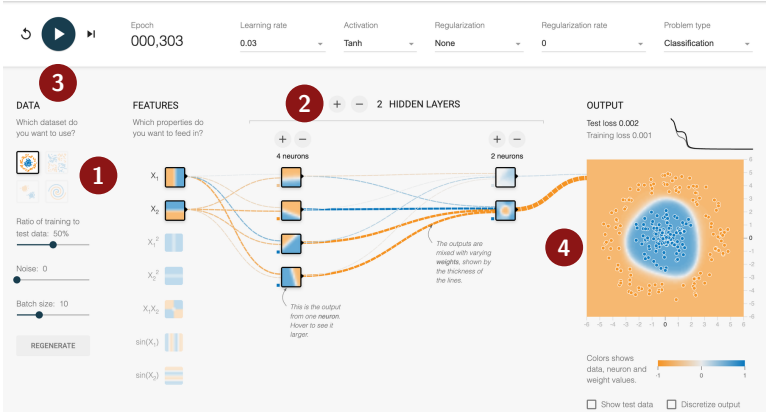
Methods

Deeper networks, better training, the transformer (2017)

- 2012: a deep network (AlexNet) won the ImageNet image contest by a wide margin.
- 2024: Nobel prizes in physics (Hopfield, Hinton) and chemistry (Hassabis and Jumper, with Baker).

Activity: training a neural network

The TensorFlow Playground



1 Pick the data. The sliders below set the share used for training and the noise.

2 + and − add or remove layers (top) and neurons (above each layer).

3 Press play to train. One epoch is one pass through the training points.

4 The background is the network's guess. Test loss is its error on new points.

playground.tensorflow.org, by Daniel Smilkov and Shan Carter. Screenshot: the circle after 300 epochs.

Activity: training a neural network

Activity devices open, in pairs

On a laptop, open playground.tensorflow.org

1 Circle



Data: the circle (top left)
Hidden layers: 1, with 1 neuron

How few neurons can separate the two colors?

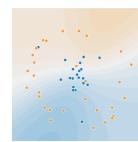
2 Spiral



Data: the spiral (bottom right)
Hidden layers: 1, with 3 neurons

Add layers and neurons. Can you get the test loss under 0.1?

3 Few noisy points



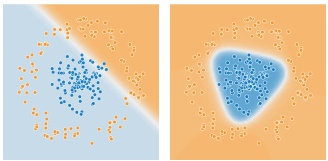
Data: the circle, noise 40, training data 10%
Hidden layers: 2 of 8 neurons

Press play and watch the loss chart. What do the two lines do?

One laptop per pair is enough. Keep the inputs at X_1 and X_2. Your points will differ a little from the pictures.

What the networks did

1 Circle



1 neuron

test loss 0.43

3 neurons

test loss 0.01

One neuron draws one straight line.
Three neurons draw a triangle.

2 Spiral

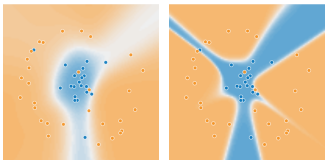


2 layers of 8

test loss 0.02

The boundary follows the two arms.

3 Few noisy points



epoch 200

test loss 0.29

epoch 2,000

test loss 0.46



Test loss fell until about epoch 200, then rose.

Prediction in policymaking

Case 1: a risk score in U.S. courts

Prediction Fails Differently for Black Defendants

	WHITE	AFRICAN AMERICAN
Labeled Higher Risk, But Didn't Re-Offend	23.5%	44.9%
Labeled Lower Risk, Yet Did Re-Offend	47.7%	28.0%

Overall, Northpointe's assessment tool correctly predicts recidivism 61 percent of the time. But blacks are almost twice as likely as whites to be labeled a higher risk but not actually re-offend. It makes the opposite mistake among whites: They are much more likely than blacks to be labeled lower risk but go on to commit other crimes.
(Source: ProPublica analysis of data from Broward County, Fla.)

ProPublica's table. "Re-offend": a new arrest within two years. "Higher risk": a score of 5 to 10 out of 10.

- COMPAS, a company's risk score (137 questions, none about race), went to Broward County, Florida, judges deciding pretrial release.
- ProPublica checked over 7,000 people arrested in 2013 and 2014.
- **The company's test:** a "higher risk" label meant about the same chance of a new arrest for Black and white defendants.
- **ProPublica's test:** among people with no new arrest, Black defendants had been labeled higher risk about twice as often (44.9% vs. 23.5%).

Source: ProPublica (Angwin et al. 2016), analysis of Broward County, Fla., data.

Case 2: a benefit-fraud model in the Netherlands

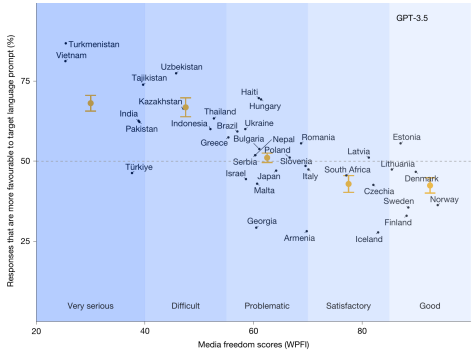


The report's cover. Illustration: Tosca van der Weerden, for Amnesty International (2021).

- From 2013, a tax-office model scored childcare-benefit claims for fraud risk.
- It learned from past claims marked correct or incorrect.
- One input was "Dutch citizenship: yes/no". Non-Dutch applicants scored higher.
- It was part of a scandal in which tens of thousands of parents were falsely accused of fraud. The cabinet resigned in January 2021.

Source: Amnesty International (2021), *Xenophobic Machines*, pp. 5–16.

Case 3: state media in chatbots' training data



GPT-3.5 in 37 countries: how often the local-language answer was rated more favorable to the country than the English one. From Figure 5.

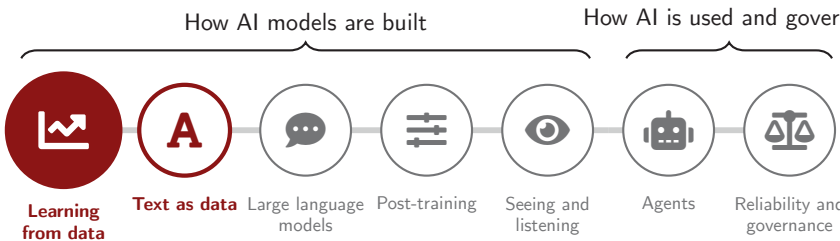
Source: Waight et al. (2026), *Nature* 655: 685–693; figure on p. 691.

- In an open training set, 3.1 million Chinese documents (1.64%) share long passages with news scripted or curated by China's state.
- After extra training on 64,000 state-scripted articles, an open model's answer to "Is China an autocracy?" went from yes to no.
- Raters judged GPT-3.5's answers about China more favorable to the government in Chinese than in English 75% of the time.
- Across 37 countries, chatbots favored the government more in the local language where the press is less free.

Discuss

1. For each case: what was predicted, from which data, and who could challenge the result?
2. The Netherlands had courts, a free press, and a parliament. Why did it still go wrong?
3. Back to today's question: when is a model that predicts well good enough for policymaking?

Next week



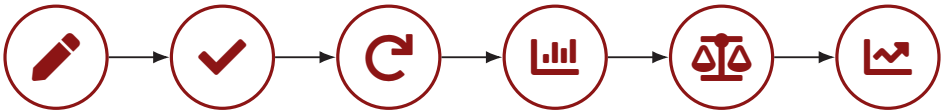
Next Wednesday, October 7

How machines read text. Project proposal due Sunday, October 11.

Backup: predicting elections with a rule

Activity: predicting elections with a rule

Activity groups of three or four, devices closed, handout on paper



Your rule can use any column except the last. Do not change it after you turn the handout over.

Training data: 1952 to 1996

Year	President's party	Growth (%)	Elected president running?	Party in power 8+ years?	President's party in the popular vote
1952	Dem.	1.0	no	yes	Lost
1956	Rep.	−1.3	yes	no	Won
1960	Rep.	0.6	no	yes	Lost
1964	Dem.	5.1	yes	no	Won
1968	Dem.	5.1	no	yes	Lost
1972	Rep.	5.9	yes	no	Won
1976	Rep.	3.8	no	yes	Lost
1980	Dem.	−3.6	yes	no	Lost
1984	Rep.	5.4	yes	no	Won
1988	Rep.	2.3	no	yes	Won
1992	Rep.	3.1	yes	yes	Lost
1996	Dem.	3.3	yes	no	Won

Fair's data. Growth: real GDP per person, first nine months of the election year, annual rate. 1976: Ford was never elected president or vice president, so he counts as "no". Johnson (1964) had been elected vice president, so he counts as "yes".

New data: 2000 to 2024

Year	President's party	Growth (%)	Elected president running?	Party in power 8+ years?	President's party in the popular vote
2000	Dem.	2.0	no	yes	
2004	Rep.	2.2	yes	no	
2008	Rep.	−1.4	no	yes	
2012	Dem.	1.2	yes	no	
2016	Dem.	1.2	no	yes	
2020	Rep.	−3.5	yes	no	
2024	Dem.	2.0	no	no	

Apply your rule to each row.

New data: the results

Year	President's party	Growth (%)	Elected president running?	Party in power 8+ years?	President's party in the popular vote
2000	Dem.	2.0	no	yes	Won
2004	Rep.	2.2	yes	no	Won
2008	Rep.	−1.4	no	yes	Lost
2012	Dem.	1.2	yes	no	Won
2016	Dem.	1.2	no	yes	Won
2020	Rep.	−3.5	yes	no	Lost
2024	Dem.	2.0	no	no	Lost

2000 and 2016: the president's party won the popular vote and lost the Electoral College.

Your rules on old and new elections



- Which group had the highest training accuracy? The highest new-data accuracy?
- Did the most complicated rule do best on new elections?

Three rules on old and new elections

	The president's party wins the popular vote if ... (otherwise it loses)	**1952 to 1996**	**2000 to 2024**
A	the elected president is running	_____ of 12	_____ of 7
B	(1) the elected president is running, the party has held the White House for under 8 years, and growth is above -2, **or** (2) growth is above 2 and below 3	_____ of 12	_____ of 7
C	the elected president is running, the party has held the White House for under 8 years, and growth is above 0	_____ of 12	_____ of 7

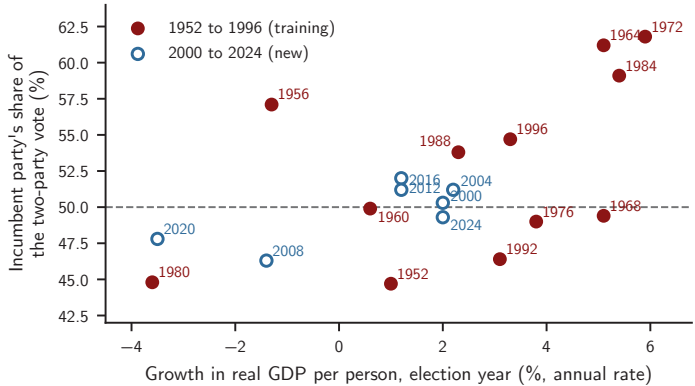
Which rule fits the old elections best? Which predicts the new ones best? Where does your own rule fit?

What exactly are we predicting?



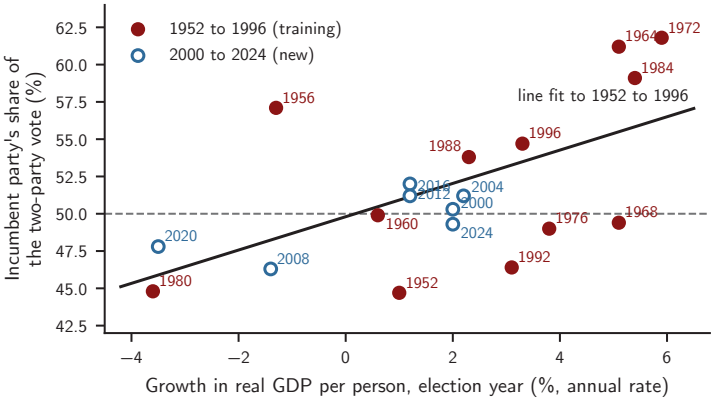
- In 2000 and 2016 the president's party won the popular vote and lost the Electoral College.
- Score rules A, B and C again on 2000 to 2024, with "won the presidency" as the outcome.
- Which outcome should a forecaster predict? Who makes that choice, and when?

From won or lost to a vote share



Draw a straight line through the 1952 to 1996 points. What vote share does it give for 2024, when growth was 2.0%? Data: Ray C. Fair (fairmodel.econ.yale.edu).

Economic growth and the presidential vote



Ray C. Fair's vote-share data (fairmodel.econ.yale.edu). The line fit to 1952 to 1996 gives 52.0% for 2024. Fair's October 30, 2024 prediction, from a model with more variables: 49.47% of the two-party vote for the Democratic candidate. Result: 49.25%.

Sources

Amnesty International (2021). *Xenophobic Machines: Discrimination through Unregulated Use of Algorithms in the Dutch Childcare Benefits Scandal*.

Angwin, J., Larson, J., Mattu, S., and Kirchner, L. (2016). "Machine Bias." *ProPublica*, May 23.

Fair, R. C. (2009). "Presidential and Congressional Vote-Share Equations." *American Journal of Political Science* 53(1): 55–72. Data and 2024 update: fairmodel.econ.yale.edu.

James, G., Witten, D., Hastie, T., Tibshirani, R., and Taylor, J. (2023). *An Introduction to Statistical Learning, with Applications in Python*. Springer. Free at statlearning.com.

Jean, N., Burke, M., Xie, M., Davis, W. M., Lobell, D. B., and Ermon, S. (2016). "Combining Satellite Imagery and Machine Learning to Predict Poverty." *Science* 353(6301): 790–794.

Smilkov, D., and Carter, S. (2016). *TensorFlow Playground*. playground.tensorflow.org (Apache License 2.0). The activity's pictures are screenshots of our own runs.

Waight, H., Yang, E., Yuan, Y., Messing, S., Roberts, M. E., Stewart, B. M., and Tucker, J. A. (2026). "State Media Control Influences Large Language Models." *Nature* 655: 685–693.

Zhang, H., and Pan, J. (2019). "CASM: A Deep-Learning Approach for Identifying Collective Action Events with Text and Image Data from Social Media." *Sociological Methodology* 49(1): 1–57.

3Blue1Brown (2017). "But what is a neural network?" YouTube.

Session 1 figures: Rungta, V. (2024). "AI Anatomy." *Chain of Thought* (Substack), May 31. Yuan, Y. (2024). *CSCI 3370 Deep Learning*, Lecture 1, slide 16, Boston College, after Fei-Fei Li and Justin Johnson.

Icons: Font Awesome Free 5 (CC BY 4.0). Video: animated with Manim Community Edition; narration by Kokoro-82M, a synthetic voice.